

		unchanged, the tension in the wire increases [B1] . Therefore, the power output of the motor increases.	
2	(a)	FBD at the top $N + W = mv^2 / r$ $N = mv^2 / r - mg$ $= 800(17)^2 / 7.2 - 800 (9.81)$ $= 24200 \text{ N}$ [Zero marks if student simply calculated centripetal force.]	 1 1
	(b)	The direction of the centripetal acceleration changes such that it is always <u>pointing towards the centre of the circle</u>. The magnitude of the centripetal acceleration decreases. Because the speed of the carriage decreases.	1 1 1
	(c) (i)	Difference in height = $h - (2 \times 7.2) = h - 14.4$ Loss in GPE = Gain in KE $mg (h - 14.4) = \frac{1}{2} m v^2$ $9.81 (h - 14.4) = \frac{1}{2} (17)^2$ $h = 29.1 \text{ m}$ [0 mark if student did not use the correct difference in height]	1 1
	(ii)	h should remain the same. [based on the calculation in (i), the h required is independent of the mass.]	1